

A Survey on the Development of Human-Centered Artificial Agents for Fighting Games

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(Survey Paper)

Abstract—Artificial intelligence methods can be used to develop artificial agents for competitive video games. These agents have the potential to be learning tools for players, as well as provide automated insights into how balanced the game is. This motivates the existence of agents whose skill is adjusted to the player. However, this also assumes the agents are good substitutes for human players. Developing “human-centered” agents, focused on the gameplay experience of human players, is a difficult problem. In the genre of fighting games, they need to account for different characteristics such as human-like adaptation and reaction time, which is an important factor in these games’ design. We survey the different dimensions by which this problem is addressed, both in commercial games and the literature. These dimensions include the performance of the agents, their human likeness and the diversity of their behaviors. We thus present the state of the art in the creation of human-centered agents, along with future directions, in which this problem can be explored further.

Index Terms—Artificial intelligence (AI), fighting game, human-like behavior, machine learning (ML).

I. INTRODUCTION

FIGHTING games are competitive two-player zero-sum video games. These are real-time games where both players act simultaneously, as opposed to turn-based play. These games are usually played between human players, but artificial opponents are also implemented. When used as a surrogate for human players, artificial opponents can be beneficial. For one, we can set up controlled learning environments in which new players can learn to play against opponents adjusted to their skill level, facilitating learnability, which is difficult to attain in these games [1]. Another benefit is for playtesting: added to the use of human testers, we can have a population of artificial agents

learn to play the game in order to assess its balance [2], [3]. This is especially important considering these games tend to be updated over time, with changes in mechanics that alter the game’s balance.

However, many approaches to creating these artificial opponents are not appropriate, in the sense that the opponents do not play like humans [4]. For instance, one major factor in this discrepancy is reaction time: many of the moves performed by each player cannot be reacted to or is difficult to do so, because of this, strategies in fighting games are heavily based on predictions of the opponent’s future moves [5]. We can think of these games as multiple consecutive trials of *Rock–Paper–Scissors* to some extent [6], [7]. Fighting games require the player to be as unpredictable as possible while predicting the opponent’s behavior patterns, similar to *Rock–Paper–Scissors* [8].

This dynamic however is either not captured by commercial games or developers are not transparent about whether it is present. Transparency in how artificial opponents work is important as players show bias when playing against them, and assume these opponents will not play intelligently or follow reasonable patterns [9]. Another factor we need to consider is the believability of the artificial opponents’ behavior. Opponents exhibiting human-like behavior make for more engaging play [10], since they mimic the way people think and so players are more inclined to be competitive against them as they would be against a human opponent.

This motivates the creation of agents that emulate the experience of playing with a human player. Throughout this work, we refer to these agents as “human-centered” agents, deriving from the term of “human-centered design.” A human-centered agent is built with a focus on the experience of the end user, taking into account aspects of human play. In this work, we are focused on the following desirable qualities.

- 1) **Performance:** The competence of the agent, i.e., its skill level, so that it can appropriately match against more experienced players [11].
- 2) **Adaptability:** Ability to adapt quickly to different opponent strategies, as players can detect exploitable patterns in the agent’s behavior [12], [13], [14], [15], [16], [17].
- 3) **Human likeness:** How credible the agent’s behavior is, such that it can be perceived to have qualities of a human player, crucial for player expectations [9], [18], [19], [20].

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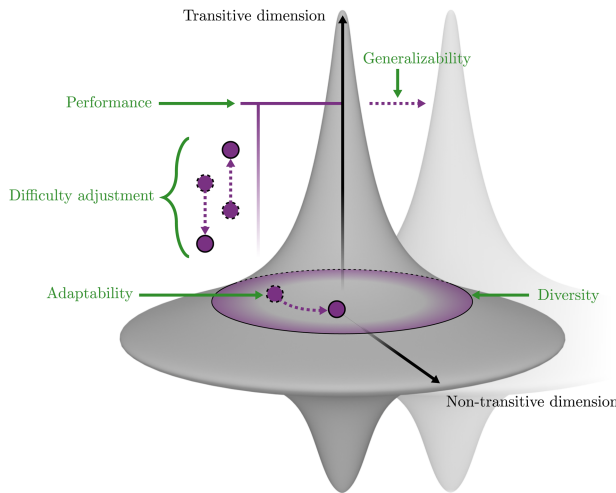


Fig. 1. Influence of human-centered qualities on the strategy space covered by artificial agents. Illustrated using the strategy space geometry of real world games proposed in [31], which resembles a spinning top. Each circle represents an agent following a particular strategy. Human likeness is not represented since it is difficult to illustrate, but its effect is a restriction on which points in the space are valid: those that represent human-like behavior.

- 4) Difficulty adjustment: Artificial agents can be beneficial to players of various skill levels, and the game should be ready to offer agents appropriate to those skill levels [11], [21], [22], [23].
- 5) Diversity: A population of agents should cover a range of strategies, which is both beneficial for training [24], [25], [26] and evaluation [22], [24], [27] of artificial agents, as well as providing a wider coverage of the game's strategy space [11], [28].
- 6) Generalizability: Ability to generalize between different playable characters, in order to economize training costs when building agents spanning a game's character roster [5], [21], generalize between opponent characters [29] and generalize between games [30].

These qualities are the result of grouping the various research goals of the works found in the literature. These are all qualities that are desirable in an artificial agent, but not all are strictly mandatory. We also argue the qualities are not equally important. Human likeness is the most important feature to strive for since without it the gameplay experience will be distant from the one had with human players, regardless of whether the agent exhibits the other qualities.

A human-centered agent should represent human players. Therefore, it should cover the space of human-like strategies. Fig. 1 presents the effect each quality has on the strategy space covered by artificial agents. The strategy space geometry is based on the one proposed in [31], and is explained further in Section VI-E. Human likeness is not illustrated, but it restricts the space to the strategies a human player would likely employ. Performance and diversity help increase the range of players effectively covered by the agent; performance is required for higher skilled players and diversity is required to challenge different playstyles. These are concerned with the space covered in the transitive and nontransitive dimensions, respectively. On

the other hand, difficulty adjustment promotes adjustment of the skill level of the agent, while adaptability promotes modification of the agent's strategy in response to the player. These two qualities are concerned with traversing the space in the transitive and nontransitive dimensions, respectively. Finally, generalizability is desirable from a technical perspective, as a general agent can be reused between character matchups, avoiding the combinatorial explosion when tackling all possible matchups [21], and applied easily to games containing different mechanics. It is represented as a different strategy space, which can correspond to a different character matchup or a different game altogether. These are all distinct qualities that may be considered when building an agent for human players, although fulfilling all of them may be difficult.

There exist various approaches to the development of artificial agents, with the most basic being artificial intelligence (AI) methods involving the manual creation of rules that determine the agent's behavior. However, these methods require extensive expert knowledge and labor. Another approach is to use machine learning (ML) methods, which involve autonomous learning of behaviors or rules to accomplish a given task. It is also possible to use evolutionary and planning algorithms, which can be simpler to adapt to the domain of fighting games since they are generally less dependent on evaluating the current game state, and are thus more problem-agnostic. They may be incorporated into other algorithms, which is often done to improve performance. For example, we can mix long-term planning with more reactive algorithms, such as reinforcement learning (RL) [32], [33], or with evolutionary algorithms [34], [35], or simply use evolutionary algorithms with artificial neural networks in general [22], [36], [37], [38].

We perform a survey on existing techniques for development of artificial opponents in fighting games, detailing what they achieve in terms of the qualities of human-centered agents. The rest of this article is organized as follows. Section II introduces the fighting game genre in more detail. Section III details the methodology used for this survey. Sections IV and V introduce existing commercial games and learning environments, respectively, detailing their advancements in the development of artificial agents for fighting games. Sections VI-A to VI-F discuss works within each quality of a human-centered agent listed earlier. Section VII presents summary notes on the contributions each method has on achieving the qualities of a human-centered agent. Finally, Section VIII concludes this article and provides directions for future research.

II. BACKGROUND

Fighting games are competitive video games in which players fight each other in an arena [39]. Famous examples of these games are those in the *Street Fighter* series. Players have a set amount of health, and the objective is to deplete the opponent's health within the time limit or remain with the health lead otherwise. Fighting games have a multitude of characters, and each player controls one of said characters. Each character has a specific set of offensive and defensive moves that the player has to use to succeed. Fig. 3 presents an example of a game state

in DareFightingICE, in which each player is using a different character.

A character's moves are performed through button presses or, in rare cases, button releases. This is the way human players interface with these games, and we can consider these button interactions to compose the most basic form of an "action space", i.e., the set of actions that the player can perform. Artificial agents can, of course, deviate from this action space and simplify it. For conciseness, whenever we mention "action" we refer to the basic action space described earlier, unless stated otherwise, e.g., as "macro actions." The complexity with which moves are performed can range from simply performing an action to having to perform a combination or sequence of actions. Moves requiring sequences of actions are usually called "special moves," and tend to be unique to each character. The actions can be subdivided into those focused on movement and those focused on attacking the opponent.

One other prominent feature of fighting games is the ability to perform "combos." A combo is a sequence of offensive moves performed with a short time interval between moves, such that if it hits the opponent it leaves no gaps for them to act back [40]. Since the release of *Street Fighter II*, combos have been a standard mechanic in fighting games, and are an important ability to learn in order to optimize play. Artificial agents should thus have the ability to perform them, which requires learning to perform sets of particular sequences of moves, unless these sequences are included in the agent's action space as actions.

A. Strategy

Fighting games have imperfect information, unlike many board games. That is, each player does not have full knowledge of the current and previous states when making a decision. Considering chess as an example, each player has access to the full state of the board at all times, which is all that is required to make appropriate decisions. This contrasts with fighting games: these are real-time games with asynchronous moves from both players, and since most of these moves are too fast for a human being to react to, players can never make instantaneous decisions based on those moves. That is, at each instant players have to make decisions without being sure of what move the opponent is doing at that exact moment, and so without knowing the whole game state [34]. As such, it requires predicting what the opponent will do. Because of this, a fair artificial agent should emulate this reaction time and thus rely on prediction as well as reaction, or else it would not be playing by the same game rules as a human player [17], [25], [41], [42].

Fighting games only have perfect information when each player can perceive the current state and act on it instantaneously, which corresponds to a naive implementation of fighting game agents. In this scenario, however, these games lose their strategic appeal. Fig. 2 shows an example of a possible situation in the fighting game *FOOTSIES* [43]. The current state at time step t can be perceived instantaneously by an artificial agent, but human players can only reliably act on state that occurred on a variable number x of time steps earlier. At $t - 4$, player two on the right side of the screen began attacking. Attacks have some

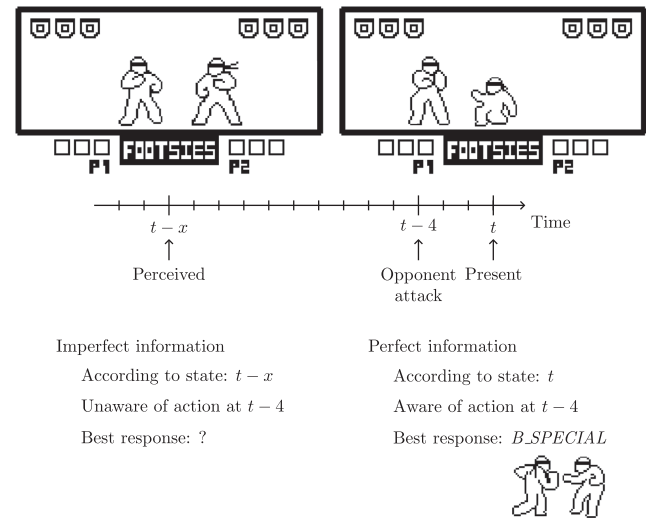


Fig. 2. Influence of perfect information on decision making in fighting games. If players are perfectly informed of all states and actions up to the present, they can perform *B_SPECIAL* in response to any attack, including itself. In that case, players should simply wait for the opponent to attack, since attacking will leave the opponent vulnerable. In this scenario, no player is incentivized to attack, stalling the game. Because human players cannot react to these attacks, this strategy is infeasible.

duration, and at time t player two is still performing the initial portion of the attack, and will only hit the player in the next time step. Informed with this knowledge, player one can perform a particular special move, internally identified as *B_SPECIAL* in the game. This move is invulnerable at the beginning and hits early, allowing player one to counter-attack player two's move. However, its main weakness is that it takes long to finish, which leaves the player vulnerable in case it misses or is defended against. If players can always react instantaneously, then they can perform this move with the guarantee that it will hit, greatly diminishing its weakness, and because of this, the game becomes trivialized. On the other hand, reacting this fast is impossible for human players, and in that case the move carries some risk. This example illustrates how fighting games are fundamentally different from the human perspective: they require players to rely on predictions of the future to compensate for the reaction time delay.

Because of the prediction-based play and nontransitivity of moves, fighting games can be thought of as multiple consecutive trials of *Rock-Paper-Scissors* [6]. Unlike *Rock-Paper-Scissors* however, the trials are not independent of each other, i.e., the outcome of one trial influences the trials performed in the future, since fighting games are prolonged in time. In addition, each move does not yield the same risk or reward, and has long-term consequences that influence the reward that that move is expected to bring in the future. Finally, since each player's moves can be executed at an arbitrary time and take some time to perform, these *Rock-Paper-Scissors* trials in fighting games are extended in time rather than being decided in an instant.

In a competitive game featuring multiple players, a Nash equilibrium is a situation, in which there is no other strategy that could be employed by each player that is better than their current

one. A Nash equilibrium exists for *Rock–Paper–Scissors*, which involves players randomly choosing any move uniformly [44]. However, fighting games are more complex than *Rock–Paper–Scissors*, and uniform random play may not constitute a Nash equilibrium. An analysis of the Nash equilibrium for fighting games has been done [7], suggesting the Nash equilibrium is indeed a mixed strategy involving stochastic play, but it is incomplete as ample player positioning and the duration of moves were not considered. Still, due to their nontransitivity, we can expect Nash equilibria for fighting games to still be composed of mixed strategies, so that they are not exploitable by being predictable.

However, as is in the *Rock–Paper–Scissors* game, if we are facing an opponent with a specific strategy outside an equilibrium, we can gain an advantage by attempting to use a counter-strategy, yielding greater return than when the equilibrium is followed [45]. In fact, human players tend to deviate from random play in order to exploit patterns in the opponent's behavior [45], [46], [47]. Even when attempting to employ stochastic strategies, human players tend to have biases that drive them to choose actions not randomly but according to some subconscious pattern [8], [31]. This means that, against a specific human player, a Nash equilibrium may not represent the optimal situation and, as such, a method that adapts to the opponent's biases will be more appropriate. Still, the objective of achieving the best performance possible is the one most explored in the literature, detailed further in Section VI-A. The objective of adapting to the current player is explored by works in dynamic difficulty adjustment (DDA), detailed further in Section VI-D.

B. Challenges

This genre poses the following challenges in implementing a human-centered artificial agent.

- 1) Decision time is standardized to 1/60th of a second, since most of these games process game state at a fixed rate of 60 frames per second.
- 2) The games are prediction-based rather than reaction-based. Human players cannot react quickly enough to many of the moves that the opponent can make, and these games are built with that in mind. As such, artificial agents should take this into account.
- 3) Specific sequences and combinations of actions yield specific moves. This increases input complexity, as action sequences require short-term planning, and action combinations increase the effective number of distinct decisions and require multiple action predictors.
- 4) An ability to adapt to the opponent's behavior quickly is paramount. As explained previously, playing optimally against a human opponent requires exploiting biases in decision making.
- 5) Combos require long-term planning of moves, which may themselves require planning at the level of actions if the moves need a sequence of actions to be executed.

As such, a human-centered agent for fighting games should exhibit consistent and efficient decision making, be prediction-based, be able to execute sequences and combinations of actions,

be quickly adaptive and perform long-term planning. The input complexity can be slightly alleviated by considering combinations and special moves to be actions themselves, although it makes the action space dependent on the character.

III. METHODOLOGY

The survey began with building a broad collection of papers related to the fighting game genre, regardless of the work contained therein, to get an overview of research in the area. This brought to our attention some papers that delved into useful concepts, such as [6], [7], even though they are not strictly related to the development of artificial agents. These papers were still important to deepen our understanding of the genre, and proved useful in contextualizing the problem. This search was performed using the Google Scholar, Scopus, and IEEE Xplore search engines. Preference was given to the most recent papers, and results were deemed relevant through analysis of their abstract.

This initial collection of papers was registered in a knowledge base and then reviewed. Whenever we found references of interest while reading the papers, we would read their abstracts and determine if they were relevant for our survey, and if so, include them in the collection. This is akin to the backward snowballing approach for systematic literature reviews [48]. In addition, whenever there was a topic of interest that we found worth researching further, we would broaden our scope and use the aforementioned search engines to find papers related to that topic. Examples of such papers are [31] and [49].

The papers were categorized according to the problem and method that they employed. The list of categories was built incrementally as new papers were reviewed and we felt more categories had to be included or others had to be excluded. The final list of problem categories was “diverse play styles,” “human likeness,” “adaptability,” “DDA,” “generalizability,” and “opponent modeling.” The final list of method categories was “RL,” “supervised learning (SL),” “planning-based,” “evolutionary algorithm,” “rule-based,” and “imitation-based.” We additionally employed two other categories: “DareFightingICE” includes papers that used the DareFightingICE/FightingICE platform, detailed in Section V-A, and “game theory/psychology” includes all papers that dealt with either game theory, game design or psychology. This categorization proved helpful, as we were able to cluster the papers according to problem-method category pairs.

The bulk of the search was performed in the last quarter of 2023. Since then, the collection of papers was incrementally updated as this survey was being written and our knowledge on the subject improved. Whenever during searches for information we found good candidates for papers to review that we missed or were published after 2023, we included them in the collection. The search was repeated in 2025 to try making sure we did not miss papers from 2024.

IV. COMMERCIAL GAMES

Almost all commercial fighting games either do not employ AI systems exhibiting the previous qualities or are not transparent

about their implementation. Of note, the 2013 release of *Killer Instinct* contains a “Shadow System,” which creates artificial opponents based on play data of human players [41]. An artificial opponent from this system has a replay bank of matches played by a specific human player, and during inference the agent chooses the moment in the replay bank that most closely resembles the current situation according to a similarity metric. This similarity metric includes heuristics that allow incorporating historical data into the current state, requiring some expert knowledge. After the moment in the bank is chosen, the agent imitates the human player’s actions from that moment forward, until a significant state change is detected.

This method is an instance of case-based reasoning, in which an AI system utilizes solutions to past problems to solve new similar problems. Although decision making is reduced to copying past human experience, the method ended up working well in practice with a positive reception from the players [41]. The agents are also easy to train, requiring few samples of human play data to exhibit seemingly meaningful and complex behaviors [41]. Because human play is copied, the agents also inherently retain human limitations, such as reaction time and execution imperfection. The method also has the added benefit of adapting to the players’ strategies as they evolve, by simply prioritizing more recent play data, without needing any further tuning.

However, the AI from the Shadow System does not explicitly perform decisions, but simply imitates other players. This means that the AI is not necessarily playing to win. If the human player it is imitating is losing on purpose, then it will lose on purpose as well. The AI does not have any notion of how good or bad each decision is, i.e., the replay moment it chooses, in terms of future reward. This allows the AI to exhibit all kinds of behaviors including noncompetitive ones, such as taunting, but not to autonomously provide insight into how balanced the game is, for instance. It also does not play with the intention of countering the player’s strategy directly, which has been found to be a limitation of case-based reasoning in the literature [50]. We want to note that case-based reasoning is not to be confused with “imitation learning,” which is a paradigm in RL focused on a similar goal, but encompasses a different set of methods, such as SL and inverse RL [51].

Another example worth detailing is *Street Fighter 6*, released in 2023. The developers designed its AI with human-centered features in mind, mainly human likeness and difficulty adjustment [52]. Their intent was to create an artificial opponent that can aid a player in learning the mechanics of the game, convey the particularities of a character, and help determine a player’s skill level before they use skill-based online matchmaking. This is a human-centered approach that aims to improve the player experience by providing agents that play similarly to human players. In addition, the agent needs to be useful for players of different skill levels. To do so, the developers incorporate three main components in the agent: abstractions for the agent’s action space, an emulation of the feeling of pressure and variable knowledge of the game. By tuning these three components, we can create agents of different difficulty levels.

Street Fighter 6’s agents do not use the basic action space where each action is a button interaction. Instead, sequences of actions are abstracted into macro actions. This is similar to the way human players make decisions; rather than thinking in terms of button presses or releases, human players decide on a move or sequence of moves to perform. Each action in an action space has information associated with it that influences the decision, such as the damage and speed of the move. The action space is different for each character, which is necessary since the moves of each character may require different sequences of actions to perform, and characters encourage different play styles. In addition, the complexity of the action space varies with the desired difficulty of the agent, with harder agents including more features of the game such as special moves. Finally, the precision of actions changes with difficulty: on easier difficulties, the agent is prone to making mistakes when executing a macro action, i.e., pressing one button instead of another, akin to the mistakes done by human players. Decision making over this action space is performed using finite-state machines. Care was taken to make the AI dynamic, i.e., rely as much as possible on the current state rather than predefined information. This allows the AI to be resilient to changes in the game’s balance, that occur over time.

The AI in *Street Fighter 6* contains artificial handicaps to emulate characteristics of human players. A pressure coefficient is calculated during the match that influences the precision with which an agent performs actions. The lower the difficulty, the more the agent is affected by pressure. Another factor that is incorporated is imperfect knowledge. The agent has three different levels of knowledge about whether it is advantageous in a particular state, i.e., whether it can act before the opponent can: does not know, knows whether it is advantageous, and knows exactly by *how much* it is advantageous, in terms of frames. For lower levels of difficulty, the agent has less knowledge about advantage, which affects decision making. Tuning these components allows modulating the difficulty of the agent in a manner that is consistent with how skill varies in human players, making it able to exhibit human-like behavior across difficulty levels.

There are other noteworthy examples that, although revealing their general approach, were not fully transparent on the solution. *Samurai Shodown* [53], released in 2019, uses deep learning (DL) for its AI. This system, similar to *Killer Instinct*’s, imitates human players and creates “Ghost” characters for each one. This allows human players to emulate playing against themselves, as well as play against artificial agents trained on other players. However, details on the system’s architecture or reception by the players have not been disclosed. *Tekken 8*, released in 2024, features a “Super Ghost Battle” mode that allows each player to build an AI that learns to imitate them, akin to the above examples. The AI utilizes RL to learn, more specifically the Q-learning algorithm [54]. However, no more details have been disclosed, and it is not clear whether the reward for the Q-learning algorithm is a reward for closely imitating the human player or for playing the game successfully, only utilizing the human play data to know what experience to use for the updates. Also in 2024, *Street Fighter 6* introduced the “V-Rival” system [55], which is similar in goal to *Killer Instinct*’s Shadow

TABLE I
COMPARISON BETWEEN AI SYSTEMS IN COMMERCIAL FIGHTING GAMES

System	Game	Year	Goals	Method	Implementation
Shadow System	<i>Killer Instinct</i>	2013	Capture the experience of playing with a human player Be resilient to game updates Create, share, and play with imitations of human players	Mimicry	Case-based reasoning
Figure Players	<i>Super Smash Bros. 4</i> <i>Super Smash Bros. Ultimate</i>	2014 2018	Create, share, and play with imitations of human players	—	—
Ghost	<i>Samurai Shodown</i>	2019	Create, share, and play with imitations of human players	—	Deep learning
In-game AI	<i>Street Fighter 6</i>	2023	Capture the experience of playing with a human player Be resilient to game updates Be tunable in order to produce different difficulty levels	Rule-based	Finite-state machines
Super Ghost Battle	<i>Tekken 8</i>	2024	Create, share, and play with imitations of human players	RL	Q-learning
V-Rival	<i>Street Fighter 6</i>	2024	Create, share, and play with imitations of human players	—	—

System. Individual player data is collected and used to train artificial agents of that player, that can then be played against by anyone. The matches with V-Rival agents are also analyzed, after which comments and recommendations for improvement are given. However, no details were given on its implementation. The *Super Smash Bros.* series of fighting games feature “Figure Players,” which are trainable AI that learn to play similarly to the human player they fight against, scanned and saved through near-field communication on amiibo toys [56]. Like the previous examples, it is not known exactly how these “Figure Players” learn.

Table I compares the aforementioned AI systems in terms of goals, method and implementation. Almost all systems aim to create agents from imitation of human players. Direct imitation of human players allows creating human-like AI, which helps bridge the gap between the experience of fighting in-game AI and other human players, useful to improve learnability. The in-game AI of *Street Fighter 6* stands out as the only one that is not based on imitation, and instead the decision making is entirely engineered by the developers. It also differs from the other examples in terms of providing different difficulty levels; the imitation-based AI can supply agents of varying skill levels if there is play data of players of different skills, whereas *Street Fighter 6*’s AI has tunable parameters. Along with the Shadow System, it was designed in such a way that game balance updates do not require extensive changes on the agents. For *Killer Instinct*’s Shadow System, the AI simply needs to be supplied new play data, while in *Street Fighter 6* the AI relies on data from the current state. For *Samurai Shodown*, while the general approach is known, it is not known which method is used, i.e., SL, RL, etc. To the best of the authors’ knowledge, there are no commercial games that employ AI that performs online learning on its own, other than learning to imitate the opponent. Overall, we can note that commercial games tend to apply methods based on imitation for building human-like AI, with *Killer Instinct* and *Street Fighter 6* being those whose method is known in greater detail.

V. LEARNING ENVIRONMENTS

There are not many fighting game environments built specifically for the development of AI, especially since fighting games are mostly commercial games. Still, we identify three platforms

offering such environments: DareFightingICE/FightingICE, DIAMBRA Arena, and FightLadder. DareFightingICE was formerly known as FightingICE prior to 2022. Hereafter, we will refer to the platform as a whole as “DareFightingICE” for conciseness, and refer to it as “FightingICE” when we specifically want to mention versions of it prior to 2022.

Table II presents a summary comparison between the identified platforms. DIAMBRA Arena and FightLadder have a large number of environments available, as they emulate existing commercial games. DIAMBRA Arena is the platform that offers the greatest variety in fighting game subgenres, with the other two only providing 2-D nontag fighting game environments. 2-D games allow movement in two dimensions, whereas 3-D games allow players to move into the foreground and background. In tag-based games, players are able to use multiple characters and switch between them on demand during the match, whereas usually players only control one at a time. While DareFightingICE endorses any kind of method, the other two platforms are mainly focused on RL. FightLadder is the environment that officially provides the largest number of pretrained sample AI, although DareFightingICE has a large catalogue of open-source agents used in its competitions. Although DIAMBRA Arena has hosted one competition and has a live leaderboard, the submitted agents are not available to the public. DareFightingICE is the only platform providing audio features, and although at one point also supported the Gym API [57], it was deprecated. In terms of modifications on the basic action space described in Section II, DareFightingICE and FightLadder discretize special moves, i.e., include the required sequence and combinations of actions as a macro action. DIAMBRA Arena and FightLadder also provide multi discrete action spaces, where movement or defensive actions are separated from offensive actions and can be performed simultaneously.

The following sections describe each platform in more detail.

A. DareFightingIce

DareFightingICE is a fighting game made for research of AI algorithms using any methodology, released in 2013. This platform was developed by the Intelligent Computer Entertainment Lab¹ at Ritsumeikan University, and is illustrated in Fig. 3.

¹<https://www.ice.ci.ritsumeikan.ac.jp/links.html>

TABLE II
COMPARISON BETWEEN LEARNING ENVIRONMENT PLATFORMS FOR THE DEVELOPMENT OF ARTIFICIAL AGENTS FOR FIGHTING GAMES

Feature	DareFightingICE	DIAMBRA Arena	FightLadder
Release year	2013	2021	2024
Number of environments	1	9	5
Environment source	Custom	Commercial	Commercial
Sub genres covered	2-D non-tag	2-D tag/non-tag and 3-D tag/non-tag	2-D non-tag
Main methods	Any	RL, imitation learning	RL
Pre-trained sample agents	3	3, across different games	5 for each screen side, for one game only
Competitions	12	1, and a continuous competition through live leaderboards	—
Participant source code	Yes, up to 2023	No	—
State representation	Structured, visual and audio	Structured and visual	Structured and visual
Action space modifications	Discretized special moves	Multi discrete	Multi discrete, discretized special moves
Interface type	Java library and Python library (gRPC)	Gymnasium API	Gym API



Fig. 3. Screenshot of a round in DareFightingICE, version 6.3.1, between the characters ZEN and LUD.

This game works like usual fighting games, with a framerate of 60 frames per second and specific action sequences activating special moves [58]. Note that these action sequences are only mandatory for human players, and artificial agents can simply execute special moves by activating a sequence of actions. Agents interface with the game through a Java library or, alternatively, a Python library, which communicates with the game through gRPC [59], although the game recently transitioned to using sockets rather than gRPC.

The Intelligent Computer Entertainment Lab hosts yearly competitions since 2013, in which participants develop AI agents for the platform to fight each other in a tournament [60]. The main goal of the competition is to advance research in the development of general AI, specifically for fighting games. A general fighting game AI should be able to adapt to and surpass unseen opponents, using any kind of character that the game offers. Since 2022, the platform changed its name from “FightingICE” to “DareFightingICE,” and the competition changed focus to the development and evaluation of sound designs appropriate for gameplay by visually impaired players. The DareFightingICE competitions have two separate sections, one for sound design development and another for agent development using sound as sensory input, which became independent competitions in 2024.

Artificial agents can perceive the game state using three distinct representations: audio, visual, or structured. Audio-based features are a recent addition that has been experimented with in the DareFightingICE competition. It has been proven that, with appropriate sound design, fighting games can be played based on sound cues alone.² Utilizing audio-based features for creating artificial agents is also an interesting problem that has not been as extensively explored as using visual-based ones. The game attempts to emulate the reaction time of human players by delaying the structured state information sent to the artificial agents by 15 frames, which corresponds to 250 ms. However, audio and visual data are not delayed.

Three playable characters are offered. The offering of three playable characters, each behaving differently, affords experimenting with generalization.

One important feature of the game is the ability for the agent to perform simulations of the game on demand. This heavily simplifies the application of algorithms dependent on knowing the environment dynamics, since they are already provided. This makes the implementation of planning algorithms, such as Monte Carlo tree search (MCTS) more straightforward, relieving the burden of learning an environment model from the agent. However, simulation is computationally expensive, so the time between time steps (which, as a reminder, is 1/60th of a second) is too short to allow extensive simulation [61], [62]. The ability to learn environment models could allow for the use of more abstract representations of the dynamics that encode information that is not only useful for the agent but is also faster to compute. Consider, for instance, temporally abstracted simulations that allow several frames to be performed at once. Still, learning such a model is an open research question [63].

Fig. 4 shows the evolution of methods used among the competitors for the FightingICE and DareFightingICE competitions over the years. We can broadly classify approaches as being in one of three main categories: rule-based, planning-based, and RL-based [64]. Note that each competitor may combine multiple

²Examples at: <https://www.ice.ci.ritsumei.ac.jp/~ftgaic/Downloadfiles/2022.pdf>

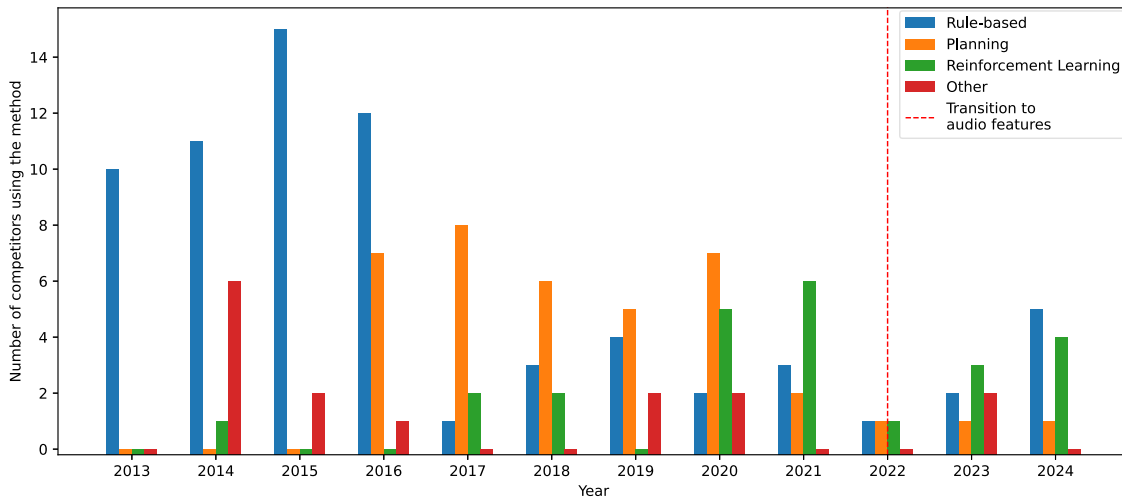


Fig. 4. Method adoption of FightingICE and DareFightingICE competitors over the years. The “other” category includes all works that neither used rules, planning or RL, such as evolutionary algorithms. From 2022, the competition transitioned from the “FightingICE” competition to the “DareFightingICE” competition. The data considers the participants’ source code and conference presentations, and includes agents developed by the organization, such as MctsAi.

methods, e.g., planning may be combined with RL, which is common. Methods that are classified into “other” are those that neither used RL, planning nor manual rules such as evolutionary algorithms.

In earlier versions of the competition, mainly rule- and heuristic-based agents were developed. Since 2016, planning using MCTS has been widely adopted among competitors, matching the year AlphaGo achieved a milestone in RL by combining RL with neural networks and MCTS-based planning [65]. MCTS is also particularly useful in fighting games since it can be executed for an arbitrary amount of time and produce partial results. Therefore, the whole limited time budget for every decision can be allotted to planning. However, it may not be a good method for a training opponent for the same reason, as it will make training occur in real time, and speeding it up sacrifices the performance of the training opponent [22].

In recent years, there has been more interest in using SL and RL for agent development. Still, most submissions to the latest competition, in 2024, were rule-based and had good performance, with four such submissions making up the top five. The preference for rule-based agents and their success may be attributed to the transition to audio features, which can still be hard to learn from using SL and RL, as they are not as extensively explored as visual and structured state features. Rule-based agents can also behave more predictably, whereas RL methods are more prone to behaving weirdly in exceptional situations [66]. Adoption of planning has seen a decline. Since 2022, the only planning-based agent was MctsAi with a very low time budget, provided by the competition organizers and the only agent not using audio-based features [60]. All submissions using SL and RL used DL methods, which can tackle large amounts of information. On the other hand, planning methods can be state-agnostic. For example, the most basic implementation of MCTS does not need to analyze the features that make up the current state; all that is required is that actions are simulated on that state. A state can be identified not by its features but by the

sequence of actions performed to reach it. As such, a possible reason for the decline is that the platform is focused on the use of audio features, and usage of methods that are agnostic to those features is not endorsed.

Various works use DareFightingICE/FightingICE as a testbed for experimentation. The platform provides a baseline MctsAi that purely plans with MCTS, made available in 2016, and two other baselines based on RL: one using a Deep Q-Network (DQN), introduced in 2018, and another using proximal policy optimization, introduced in 2022. These AIs, mainly the one based on pure MCTS, are frequently used to comparatively evaluate performance both in the competitions and the literature.

B. DIAMBRA Arena

DIAMBRA is an ecosystem comprising two platforms for development of and competition among AI, respectively, focused on RL. Although RL is the focus, the platform also provides functionality supporting imitation learning. DIAMBRA Arena is the development platform, allowing development of agents in competitive video games for research, competition and experimentation [67]. The project has been hosted on GitHub since 2021, and started supporting tournaments on the same year. It provides environments by emulating various commercial games, with the current selection featuring fighting games only. This selection is presented in Fig. 5. The supported games cover different subgenres of fighting games, providing a great variety in game mechanics. Agents interact with the environments through the Gymnasium API [68]. The platform already offers various wrappers and utilities for this API to support different use cases, such as imitation learning.

The DIAMBRA competition platform hosts a continuous competition in which competitors can asynchronously submit agents to the platform as a solution for one of the environments. The platform calculates a score for each submission, and a live leaderboard presents all submissions made to each environment



Fig. 5. Selection of commercial games available as environments in the DIAMBRA Arena platform. The current selection is diverse, covering different fighting game subgenres. Note that the versions of *Marvel versus Capcom: Clash of Super Heroes* and *X-Men versus Street Fighter* available on the platform do support tag-based fighting, contrary to the version of those games released for the *Playstation* console.

thus far, ranked in terms of score. The score is calculated by running the submitted agent for a single episode, and is a function of the cumulative reward and the game difficulty. The difficulty is specified when the agent is submitted, and can be either “easy,” “medium,” or “hard,” with each subsequent level increasing the score obtained for the same amount of cumulative reward. The score-amplifying effect of each difficulty is increased the greater the cumulative reward. When submitting, the user uploads their agent to a private repository, to which the platform has access through an access token. This means that, unlike the competitions in DareFightingICE, the participant’s agents or source code are not accessible. Instead, only statistics and video footage of each submission are available. In terms of synchronous tournaments, DIAMBRA hosted its first tournament, RLZ tournament, in 2021. Similarly, the participants’ agents and source code are not accessible.

Artificial agents perceive the game state using visual data extracted directly from the game screen, along with some structured features that can be inferred from this visual data. This not only means that the structured features are not strictly necessary to interact in the environments, but also that they do not contain any information hidden from visual perception. The structured data are not extensive, as it does not contain information, such as

the players’ positions or their current move, which is useful for determining which decisions are appropriate for some state. The intent is to mimic the way human players perceive these games as closely as possible, by focusing on control through visual features. Contrary to DareFightingICE, the structured data are not delayed.

The agent can perform actions in different ways. The platform already discretizes all combinations of movement actions that cover the cardinal and ordinal directions, so other combinations are not necessary. For the attack actions, the platform optionally allows discretizing action combinations, required for some games. Finally, it is possible to perform movement and attack actions either in mutual exclusion or at the same time. This means that the agent can either only perform a movement or an attack action at every frame, or that they can perform both at every frame. The latter is closer to the way humans play these games.

Each game provides a multitude of characters. This is useful for testing generalization, as we can evaluate generalization between games, between characters used by the agent and between characters used by the opponent. In addition, the games provide two modes of play: single-player and multiplayer. The multiplayer mode is similar to how DareFightingICE works, in which two agents/human players fight against each other in isolated games. The single-player mode on the other hand is focused on the agent fighting a sequence of opponents controlled by the AI already implemented in the game. The difficulty rises as the agent advances in the mode. Since agents are matched with opponents of increasing strength, single-player has the potential to be leveraged for curriculum learning.

The platform also provides support for multiple modes of training. Human-in-the-loop training can be performed by taking manual control of the agent and playing against the agent itself during training. It also contains environment wrappers for recording and playing experience, useful for imitation learning.

C. FightLadder

FightLadder is a benchmark focused on competitive multi-agent RL [30]. Similarly to DIAMBRA Arena, it emulates various commercial games and provides them as learning environments, which can be interacted with through the Gym API [57]. It supports five commercial games: *Street Fighter II*, *Street Fighter III*, *Fatal Fury 2*, *Mortal Kombat*, and *The King of Fighters '97*. Like the games supported by DIAMBRA Arena, they contain a multitude of characters and game modes. The games support single-player and multiplayer, as well as team mode for *The King of Fighters '97*, in which each player can use multiple characters in a match. Unlike DIAMBRA Arena however, the current selection of games does not cover a large diversity of subgenres, with all games being 2-D nontag fighting games.

The goal of the platform is to provide complex and computationally efficient competitive environments for benchmarking AI based on multiagent RL, which there is a lack of. Due to the lack of unified criteria for evaluating competitive agents, the authors propose four different evaluation criteria. The most basic, and most used in the literature, is using a measure of performance

against the built-in AI of the respective environment, which commercial games tend to provide. This measure of performance can be, for instance, win rate. Elo ratings were also proposed, which are appropriate when a large pool of opponents is available. These two criteria depend on preexisting agents. The authors also define performance in terms of robustness against exploits by specialized agents. These “exploiter” agents are trained solely to play against the agent to be evaluated, with the objective of overfitting to the agent at the detriment of generalizability. The goal is to measure the transitive strength of the agent’s strategy. This criterium contrasts with the previous two in that the measurement is dependent on the training of other agents. This means that the metric is limited in how good of an approximation the exploiter agent is of the best response against the agent, which depends on the algorithm used for the exploiters. Finally, to complement inter-agent evaluation, the authors also propose measuring performance against human players. The advantage of playing manually against the agent is that we can directly influence the game state and it is easier to understand the agent’s flaws. However, this analysis is opinionated and heavily dependent of the human players used for evaluation. None of the proposed metrics is perfect on its own, but the combined usage of all of them can bring greater insight into the agent’s performance. The platform has features that support evaluation through these metrics.

The platform focuses on visual features, but agents can also use structured data. This structured data can be used to augment the state or to debug the agent. The action space is a simplification of the basic action space where each action is a button press/release. Redundant action combinations are removed, and actions are separated into two sets: movement or defensive actions, and offensive actions. This is similar to how DIAMBRA Arena separates actions between movement and attack actions. Special moves are also discretized, similarly to DareFightingICE.

This is a relatively recent platform that is yet to be explored thoroughly.

VI. HUMAN-CENTERED QUALITIES

In the following sections, we summarize the main contributions for each quality of a human-centered agent, gathered from the literature review.

A. Performance

Most works in the development of artificial agents for fighting games tend to focus on performance rather than create agents for play against human players. For two-player zero-sum games, performance is hard to quantify in absolute terms. The evaluation metrics that are used instead are comparative, one example being Elo. These metrics are calculated by having developed agents match against other players, both artificial and human. Table III presents the different kinds of performance evaluation methods employed by various works. Health score is mostly used in the DareFightingICE competition, and is computed as

$$\frac{h_{\text{self}}}{h_{\text{self}} + h_{\text{opp}}} \times a \quad (1)$$

TABLE III
ADOPTION OF PERFORMANCE EVALUATION METHODOLOGIES IN THE LITERATURE

Metric	Applied to	Reference
Win rate	Built-in AI	[70], [71]
	MctsAi	[34], [35], [72], [73]
	Past DareFightingICE competitors	[22], [32], [33], [61], [69], [74]–[79]
	Human players	[25], [30], [73], [80]
Health score	MctsAi	[72]
	Past DareFightingICE competitors	[13], [27], [36], [66], [74], [78], [81]–[86]
Elo rating	Built-in AI	[30]
	Past DareFightingICE competitors	[26]
	Custom agent	[13]
	Own agents	[30]
SDR score	Past DareFightingICE competitors	[24]
Damage dealt	Human players	[14]
	Static opponent	[87]
Average reward	Built-in AI	[67], [80]
	Rule-based AI	[88]
Health difference	MctsAi	[89], [90]
	Past DareFightingICE competitors	[34], [69]
Ranking change	Past DareFightingICE competitors	[62]
K/D ratio	Human players	[5]
Exploitability	Own agents	[30]
Number of hits	Past DareFightingICE competitors	[37]

where a is a multiplicative factor, usually 1000, h_{self} is the health of the agent, and h_{opp} is the health of the opponent. Some works use a slight variation in which the ratio is on damage rather than health, in which case the numerator is on the damage dealt to the opponent, but we treat it as the same metric since they end up representing the same concept. SDR score is computed as

$$\frac{w \times \Delta h}{\Delta t} \quad (2)$$

where w is win rate, Δh is the health difference between both players, and Δt is the elapsed time. Compared with health score, this metric has the advantage that time is incorporated, which values the ability of agents to win quickly. This factor was likely not included in DareFightingICE since the platform already used to host a separate competition for speedrunning, in which agents try beat the opponent as fast as possible.

Although win rate and health score were the most used metrics, there is an opportunity to explore other metrics. For instance, exploitability can be used to test robustness, which can help determine the agent’s transitive strength and, therefore, its performance. However, exploitability depends greatly on how the exploiter agents are trained, and due to the metric’s recency it likely needs more investigation into reliable methods for producing exploiters. For example, the method in [69] can be used to build exploiter agents, as it is meant to find strategies in the Nash Equilibrium for fighting games. However, it is currently limited in that it does not consider reaction time, which in Section II-A was shown to be an important factor to consider. Elo rating is

also a popular metric used in the context of population-based training, and has not been used extensively in fighting games. The advantage of Elo over other metrics, such as win rate, is that it is a measure of skill that does not depend on the pairing between players, allowing easier performance comparisons.

Due to the heterogeneity in evaluation methodologies, it is difficult to compare works in terms of performance. Although most works use similar metrics, such as win rate and health score, they evaluate them on different sets of opponents.

To perform the evaluations, the artificial opponents used for testing tend to be the ones already provided in those games, which for commercial games are closed-source. For DareFightingICE the opponents used for comparison tend to be the submissions to past DareFightingICE competitions. In the case of DareFightingICE however using a large set of opponents was shown to be required, and ranking within the competitions is not directly indicative of the competitors' performance [24], [32], [61]. This suggests that there is still much nontransitivity in the strategy space covered by current agents, and as such using many test opponents is required to get a better grasp of agent strength.

Different approaches to improve the performance of artificial agents were used. Stateless agents cannot perform action sequences by themselves on the basic action space, unless the state incorporates historic information [29], [70], [74]. Another possibility is, for the case of special moves, to include the required action sequences and combinations as macro actions in the action space [91]. Since it is possible to perform multiple actions simultaneously, some works tackled actions combinations by treating them as macro actions as well [29], or by having the agent output multiple actions [71].

Some works attempted to improve the agent's ability to perform combos, by including the action history in the state [29] or through planning [33]. However, performing extended combos proves to be difficult, especially in RL as it requires a large amount of exploration [29].

Many works attempted self-play [24], [92], and it was shown to be crucial to improve performance [5], [21], [25]. It was found that naive self-play, i.e., picking the latest agent snapshots, performed worse than randomly picking past agent versions [92]. One possible reason for this result is that picking old versions of the agent provides more diversity, which promotes generalization. It is important to diversify the pool, either by including different opponents [24], using agents with different architectures [21] or including different configurations of the agent, e.g., different reward functions [25], [70]. Firoiu et al. [5] found self-play greatly helped with generalization, which was necessary to perform well against human players. However, the agent could still be exploited and perform odd behavior under certain conditions.

Frame skipping was also employed, in order to remove extreme action granularity and have the agent make decisions at a higher level of temporal abstraction [25]. Another trick employed is to modify the state such that the agent is always facing the same way, i.e., left or right [29]. In these games, it is possible for players to switch sides with the opponent, and since players always face the opponent, they will be facing the opposite side when that switch happens. This brought a

performance increase due to the reduced training data imbalance, since we manually perform generalization between states that are semantically the same. This issue can also be tackled at the level of actions, as in [70]. Finally, action masking can be employed [26], [73], since in these games the set of available actions depends on the current state.

Current agents still struggle to maintain good performance under an emulation of reaction time or state/action delay [80], [84]. In [80], it was found that tackling the delay in full, rather than partially, was necessary to obtain good performance. However, the agent was correcting the state/action delay by predicting the future state, and it impacted performance greatly when predicting the full delay, so much so that the agent could not act in real-time.

B. Adaptability

A major problem in creating efficient artificial opponents for human players in fighting games is ensuring a fast enough adaptation to the player's behavior [14]. That is, the artificial agent should change its strategy in response to the opponent's perceived strategy, such that it would be a win-maximizing response. Unless the agents employ a strategy at a Nash Equilibrium, agents cannot be static, or else they are prone to exploitation [32]. As explained in Section II, optimizing play against human players requires adapting to their play style, and not only playing randomly. Adaptation to the player has to be done quickly enough, and not require an extensive number of matches. However, the adaptation should not be too abrupt so as not to cause disorientation to the players, which can occur with ensemble agents [78]. Overall, the literature currently lacks research in this area.

C. Human Likeness

Artificial agents developed for these games tend to behave in a way that human players do not, giving away their artificiality [19]. This is due to the complex decision making and control that human players have, which may even be influenced by emotions, and therefore is hard for agents to emulate. As a result, the players' experience is often diminished due to the lack of believability in the opponent's behavior [18]. This is an imperative concern for fighting games since human players assume that the opponent has exploitable biases in their play style that can be used to their advantage. If it is clear that the opponent is artificial, human players will not think to play in this pattern-seeking way [9], diminishing engagement.

We can consider two approaches to this problem: either the artificial agents are constrained in ways similar to human players which should result in exhibiting similar behavior, or the agents directly attempt to imitate human behavior. Examples of constraint-based approaches are emulation of human reaction time [58], [80] and emulation of pressure or knowledge imperfection [52]. Other than that, current work, including commercial games, tends to focus on the second approach of imitating human behavior.

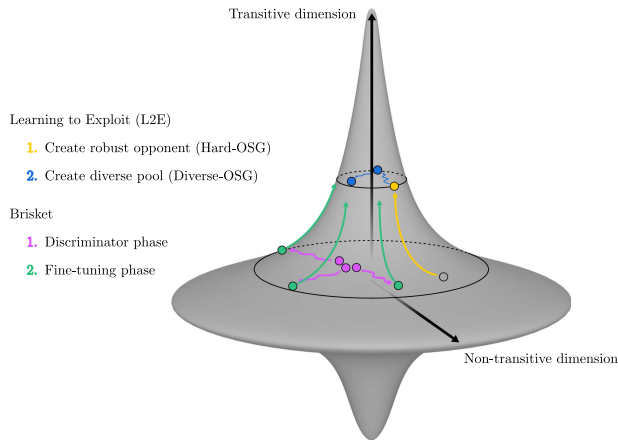


Fig. 6. Strategy space geometry of real world games proposed in [31]. The agent generation strategies in Brisket [28] and L2E [102] are illustrated using this geometry.

Once we have an approach, how do we evaluate “human likeness”? One test seen in some works is the Turing test, alternatively called the “imitation game,” in which human evaluators attempt to determine whether a given entity is human or artificial merely by its actions on a particular task [3], [93], [94]. For instance, in one-on-one competitive games, video footage of the artificial agent playing against a human or another artificial opponent is shown to human evaluators. The evaluators have to specify whether each, or one of, the players is human or not. The usefulness of this test is currently a point of controversy [95], but it seems valid since the main objective is to develop an artificial agent behaving similarly to a human, and so the ability of an agent to pass the Turing test directly represents this goal. This is in contrast to the broad original objective of Turing tests, which is to evaluate whether an artificial agent can think intelligently.

Achieving human likeness often conflicts with the pursuit of other qualities. For instance, trying to perform DDA can make the agent have uncompetitive behavior, such as running away when it is performing well, which is unrealistic [96], [97]. This requires the agent to at least avoid strategies that lie below the largest disc in Fig. 6, which correspond to strategies that purposefully attempt to lose. Another example is training using RL. Two works found that RL, while able to improve the agent’s performance, usually comes at a cost of introducing artificial behavior [94], [98]. These works introduced a preliminary behavior bias on the agent such that it is prone to producing human-like behavior. This is possible, for example, by using imitation learning with SL, which was shown to be apt for imitating human behavior [20], [98]. Still, the RL algorithms deviated from this bias. Some RL algorithms were found to produce more human-like behavior than others, more specifically policy gradient in comparison with value-based methods, such as Q-learning [5]. One possible factor for this discrepancy is that policy gradient methods directly dictate the policy’s stochasticity, whereas in value-based methods it is incorporated manually for the purposes of exploration, and as such contains artificial stochasticity. Still, RL is far from imitation learning in terms of human likeness [20].

D. Difficulty Adjustment

In video games, providing a fair and engaging challenge to players can be done in two main ways: either by having the player choose a difficulty preset according to their perceived skill, or by automatically adjusting game parameters so that adequate difficulty is presented. This automatic process is called DDA, and it is meant to enhance the end users’ experience [99]. In this case, we are interested in adjusting the artificial opponent the player is playing against. Usually, estimations of the player’s performance are made and parameters are adjusted according to the perceived skill. As such, DDA is also concerned with downgrading the performance of the artificial opponent so that they better match the skill of the human player. In this section, we focus on works that try to implement DDA in fighting games, as that is the method explored in the literature.

Most works that explored DDA in fighting games did so for FightingICE. DDA in fighting games was employed by minimizing absolute health difference between both players [12], [15], [89], [96], [100], excepting one work [101]. Small health differences can suggest that the players are equally matched, but this objective may still produce unnatural or undesired behavior, such as stalling when the agent has a large health lead [15]. Relaxing this objective can produce more proactive agents, rather than reactive, while still keeping the same difficulty adjustment performance [15].

Cherukuri and Glavin [89] followed a different approach, in which the different difficulties correspond to different snapshots of the agent throughout training. During a match, the agent switches between snapshots according to the difference in skill, which is measured as the difference of both players’ health. However, this method has the disadvantage that the easiest skill level corresponds to an untrained agent. This means that the agent acts randomly, which is not engaging to play against as it is unpredictable [3].

E. Diversity

Play style can be defined as the way someone or something plays a game, characterized by the strategies that they employ. For games, especially complex ones, a particular skill level is not defined by a single strategy. Fig. 6 illustrates the geometry of the strategy space in real-world games proposed in [31]. The strategy space of real-world games tends to resemble a spinning top, in the sense that the vertical axis encodes the strategy’s transitive strength and the width at a particular point in the vertical axis represents the nontransitive dimension, i.e., the breadth of distinct strategies of the respective strength. At the topmost tip of the spinning top we have the Nash equilibria, and the bottom tip contains strategies that purposefully attempt to lose. As such, it is natural that each player has their own play style, defining the way they approach the game, with this being especially true on lower levels of skill. We use this geometry to illustrate our points.

It is worth noting that the geometry of Fig. 6 was proposed considering *symmetric* games, i.e., games in which any two players have access to the same conditions. For example, turn-based games are asymmetric because one player has to start first,

and the other can only act afterward. Fighting games are also asymmetric unless both players are utilizing the same character. The authors include asymmetric games in their analysis by considering each player to play once in the other player's conditions, which in the case of fighting games means two matches are performed where each player plays once with their character and then again with their opponent's character.

This is an important formalism especially when considering population-based training methods that are often used to achieve high performance in competitive environments by artificial agents. In these competitive environments, in which the artificial agent is playing against other adversaries, we need to supply a good opponent pool such that we improve the agent by increasing their transitive strength. However, we may find that when using an opponent pool populated by naive self-play, for instance, the agent is just navigating along the nontransitive dimension, learning policies that merely form strategy cycles during training and not improving overall. Different self-play strategies have been adopted to attempt to avoid this, such as populating the opponent pool with "exploiter" agents that are focused on exploiting the main agent's weaknesses rather than trying to be robust against many opponents [103]. Essentially, we want to avoid traversing the nontransitive dimension and traverse the transitive one as much as possible. One theoretical result is that, if the population size is representative enough of the number of strategies of a given transitive strength, then we can guarantee improvement in terms of transitive strength. Moreover, we can note that nontransitive complexity lowers as the agent gets stronger, assuming the agent is beyond the largest disc of the spinning top. As such, there is interest in properly initializing the opponent pool so that the agent gets to a higher transitive strength as quickly as possible, e.g., by initializing the population with highly skilled opponents. In the FightingICE competition, for instance, since 2020 it has been frequent for competitors to use submissions from past years as training opponents in self-play [24].

We can additionally induce that, to create engaging play in fighting games, it is not enough to just create a single opponent. There is interest in having a population of agents, each exhibiting diverse behaviors that better represent the game's strategy space. This way, human players have to figure out and better adapt to various play styles of the artificial opponents, at any skill level. In addition, given the spinning top structure, some skill levels exhibit a greater set of strategies, so the population size can vary among skill levels.

Not many works in the literature explore this area. Of note, Halina and Guzdial [28] developed Brisket, a method for creating a population of agents with diverse behaviors for FightingICE. Training is separated into two phases, where in the first phase agents merely learn to exhibit diverse behaviors between each other, whether they be good or bad for performance at the game. Then, with behavioral bias set from the previous phase, the second phase focuses on learning how to play the game, using DQN with reduced learning rate to avoid radical changes in behavior. These two phases can be thought of as the first one focusing on traversing the game's spinning top space in the nontransitive dimension, while the second phase focuses on

traversing the transitive one. This behavior diversity is achieved using an adaptation of the diversity is all you need (DIAYN) method [104], which is a method for allowing an agent to learn a fixed set of skills that allow it to traverse distinct portions of the state space. In Brisket, instead of having diversity between skills in a single agent, we have diversity between different agents in an agent pool. As such, whereas DIAYN is focused on within-agent diversity, Brisket is focused on between-agent diversity. The developed agents were diverse, contrasting with the human-authored baselines developed through reward shaping, which converged to the same local maxima. At the same time, all agents exhibited the same performance, which was also superior to the baselines' performance.

Fig. 6 illustrates the Brisket population generation algorithm, as well as the learning to exploit (L2E) framework for comparison [102]. Although not applied to fighting games, the L2E work proposes an opponent strategy generation (OSG) algorithm similar to Brisket's, specifically for the generation of diverse training opponents. The algorithm is separated into two phases, but with inverted goals: in the first phase (Hard-OSG), a single randomly initialized opponent is optimized to be robust against the current agent; in the next phase (Diverse-OSG), a diverse opponent pool is iteratively created by creating opponents similarly to the first phase, but with the additional constraint of being dissimilar to all opponents in the pool so far. Diversity is enforced by ensuring opponents have behavior that produces distinct trajectories during gameplay. Through ablation tests, the authors found that utilizing both optimization phases (optimizing for robustness and diversity) helps increase the agent's expected return, with diversity being a large contributing factor. This work highlights the need to have not only a robust but also a diverse set of opponents during training, especially important when employing self-play. We find this work worth mentioning as a contrast to Brisket, and highlight how population generation for fighting games can still be explored further as it is not a clear science.

F. Generalizability

Considering the variety of characters present in these games, it is desirable that agents have a degree of generalization. Agents should be able to transfer skills not only between playable characters but also between opponent characters. This can be performed through transfer learning. Transfer learning denotes the process of an ML model, trained for a given task, to be trained for a different task such that the bias brought by the first task improves learning and performance on the second task. This is desirable when the two tasks are only slightly different and makes training much more efficient. In fighting games, we can think of transfer learning as the ability for an agent, trained on playing as a specific character, to be able to play as another character, but without requiring the onerous training as was needed for the first character since some skills should be transferable. Or, similarly, an agent trained to play *against* a character or opponent being able to efficiently learn how to play against another one. This is especially pertinent when games have a large character roster [21]. As such, transfer

TABLE IV
WORKS RELATED WITH EACH PAIR OF AGENT-DEVELOPMENT TECHNIQUE AND QUALITY OF A HUMAN-CENTERED AGENT

Technique	Performance	Adaptability	Human likeness	Difficulty adjustment	Diversity	Generalizability
Rule-based	[13], [14], [78], [83], [84], [115]	[13], [14], [78]	[14], [52], [115], [116]	[52]	[52]	—
Mimicry	[14], [106]	[14], [41]	[14], [41], [50]	—	—	—
SL	[34], [36]–[38], [77], [81], [82], [84], [109]	—	[3], [20], [116]	[23], [101]	—	—
Planning	[4], [27], [32]–[36], [61], [62], [66], [73], [77], [79], [81], [85], [86], [90], [109], [110], [115]	[33]	[4], [96], [115]	[15], [96], [101], [112]	[111]	—
Evolutionary algorithms	[22], [33]–[38], [40], [66], [75], [83], [90], [110]	[33]	[19], [114]	[113]	[114]	—
Reinforcement learning	[5], [12], [22], [24]–[30], [32], [34], [37], [66], [67], [69]–[74], [76], [80], [82], [87]–[89], [91], [94], [117], [118]	[16], [17], [29], [105]	[5], [20], [21], [24], [29], [80], [94]	[12], [21], [23], [89], [97], [100], [112], [113]	[28], [30], [70], [94]	[5], [21], [29], [30], [92], [97]

learning has the potential to greatly reduce the time taken to train artificial opponents. To facilitate development and training, we may also be interested in having agents that are generalizable across games.

Transfer learning between player characters [5], [21] and opponent characters [29] was shown to reduce training time greatly. Curiously, in [5], the learning efficiency for each character largely corroborated the human players' perception of learning difficulty for those characters.

In terms of performance, Kim et al. [105] found that representation transfer can make adaptation to different environments easier, but can impact performance in the long-term. That is, the agent can be biased during the adaptation.

To improve generalizability, Pathak et al. [21] removed identifiers, such as the move each player is performing, from the state features. As such, only quantitative information was kept, such as the players' position and health. Although this reduced performance for specific matchups, excluding it allowed agents to exhibit more consistent behavior throughout matchups. This is because identifiers are too discriminative of each situation, and can make it difficult to generalize.

VII. DISCUSSION

In this section, we summarize the work on the main method groups identified during our survey. We present the general conclusions taken for each method group, with respect to the qualities of a human-centered agent we described previously. Table IV presents the works we identified that relate to each method-quality pair, condensing 77 works in total. We can notice at a glance the lack of study for some method-quality pairs, such as rule-based methods for generalizability.

Beyond the rule-based, planning and RL methods prominent in the DareFightingICE competitions, we identified three others:

mimicry methods aimed at imitating the behavior of a player, SL methods used either for building a policy or for opponent modeling, and evolutionary algorithms, used for multiple purposes.

We found rule-based agents mostly in the DareFightingICE competitions; not many works in the literature exist exploring this methodology. The works we studied used either dynamic scripting [13], [14] or a mixture of experts [78], with one of them winning first place in the FightingICE competition [13]. In terms of adaptability, the mixture of experts method can have trouble adapting to the opponent since the internal agents are usually made static [78]. Dynamic scripting on the other hand has found more success, with the agent being capable of adapting within a round [13]. In dynamic scripting, the agent's behavior script is made specifically for its parts to be adjusted, and this increased flexibility allows for better adaptability.

Works based on mimicry are focused on imitating another player's behavior rather than performance. Methods include case-based reasoning [14], [41], [50] and unsupervised learning [106]. At their essence, these methods use a dataset of human play data to determine which actions to perform at the current state. One work decided to implement "skepticism" [14], which allows the agent to opt for not imitating actions, which it deems to not be useful to achieve its goal. Skepticism not only gives the agent the chance to surpass the imitated player in terms of performance, but it was also found to increase human likeness. This is a concept that can be useful to apply for other methods. The agent's actions need to span the whole action set, including the action of doing nothing, as waiting is valid human behavior [50]. For adaptability, case-based reasoning needs to consider the most recent cases and discard or give less weight to older ones [41].

SL is usually used for imitation learning [20], [37], [38] or opponent modeling [34], [36], [107], [108], [109]. In terms of performance, imitation learning is limited since the agent can

only, at best, match the performance of the player it is imitating. Most of the surveyed applications of SL either used offline learning [20], [37], [38], [107], [108] or performed learning but only between game rounds [34]. To perform difficulty adjustment, SL has been used to classify the affective state of the player to support a MCTS agent targeting that state [101]. Imitation learning with SL proves successful in achieving human-like behavior, when compared with RL [20].

Planning is usually not performed in isolation. It incorporates other methods, such as opponent modeling or RL, in order to improve the planning process, and has faced success when evaluated with opponents from the DareFightingICE competitions. For instance, the reference MctsAi in DareFightingICE assumes a random opponent during simulation [86], so some works utilized opponent modeling to make simulations more accurate [27], [34], [36], [61], [85]. On the other hand, the action set considered during planning can also be shrunk by only considering the best candidate actions, through RL [27], [32] or evolutionary algorithms [35], [110]. Planning has also been augmented through selection of hyperparameters with evolutionary algorithms [90], by including rule-based behavior to handle specific situations [62], [85], and approximation of the state evaluation function [77]. In terms of adaptation, the surveyed planning methods are dependent on the augmentation method used, e.g., the efficiency with which the opponent model adapts to the opponent [34]. Pure planning methods have still not explored believability extensively, and are yet to produce human-like behavior [4]. The planning goal can be used to define different play styles, e.g., by considering player distance [111], and for difficulty adjustment, by targeting similar health values [15], [96], [112] or affective states [101].

Evolutionary algorithms, such as genetic algorithms, are used for multiple purposes. They are used to augment planning [33], [34], [35], [66], [90], build neural networks [22], [36], [37], [38] and find combos [19], [40], [110]. Given their nature, evolutionary algorithms are not fit for adaptation unless used alongside other methods, since the iterative evolutionary process to build candidate individuals takes time. Even then, the approaches that combined evolutionary algorithms with other methods exhibited either no adaptation at all due to the generated policy or dataset being static [37], [66], [110], [113], or the methods did not learn fully online [34]. One work attempted to build a diverse population, but the agents had poor performance [114].

RL has been widely explored in the literature. It can still be difficult to apply in practice, with mixed success in the DareFightingICE competitions and against human players. Separating the reward function into subgoals can yield more stable performance [74], and using completely separate predictors for those subgoals can be even more effective [72]. To match the skill of human players, self-play was found to be an important training method for RL, either using only training snapshots [5], [25], [92] or including premade agents [24]. The methods applied in practice are sample inefficient [29], [32], [94], taking hours to train, which limits adaptability. Still, representation transfer has shown to help increase training speed [105]. Human likeness is usually implemented through constraints on the agent's behavior, either by imposing an observation or action delay emulating the reaction time of human beings [21], [24], [80],

or by restricting the agent's exploration process to avoid weird behavior [4], [94]. To handle reaction time delay, prediction of the current frame for observation delay [24], necessary for DareFightingICE, and of the future frame for action delay [21], [80] have been studied. Reward shaping has been used both to restrict the agent's behavior to be more human-like [29], and to build diverse agents [94]. Diversity can also be enforced within a population of agents through a discriminatory procedure [28]. For difficulty adjustment, we can save various snapshots of the agent throughout training and associate training time with agent skill [89]. Alternatively, we can have the agent purposefully choose suboptimal actions [100] or use custom reward so as to minimize the absolute health difference between players [12]. Finally, generalizability has been studied for RL only, and it was shown that transfer learning can be applied both between playable characters [5], [21] and opponent characters [29]. However, having an agent learn multiple playable characters at the same time is still difficult, and it is recommended to either have each agent focus on a particular character [5] or drop discrete information, such as identifiers, in the state [21].

VIII. CONCLUSION

In this work, we motivate the existence of human-centered agents for fighting games. Human-centered agents are artificial agents built with the human-AI gameplay experience in mind, providing an experience similar to playing with human players while being adjusted to the player's profile. We surveyed the literature for agent development methodologies for fighting games, as well as learning environments on which those agents can be developed, and presented their contributions with respect to each quality of a human-centered agent.

Most commercial fighting games either do not reveal their approach to agent development, or employ methods based on imitation of other players. Some works in the literature develop agents within commercial fighting games, often through emulation. This process is simplified in DIAMBRA Arena and Fight-Ladder, platforms containing various learning environments for RL-based AI, with each environment emulating a commercial fighting game. Another learning environment is DareFightingICE, which offers a game purposefully made for research and competition of AI.

Despite their straightforward approach, rule-based agents are able to be performant in some situations where automated learning approaches such as SL and RL fail. However, they are also less generally applicable since they have to be manually built. RL methods for fighting games are shown to be successful, even able to beat expert human opponents, but can still show odd behavior sometimes. Evolutionary and planning algorithms are often mixed with other methods, such as RL, to further improve performance.

There still has not been extensive work on the efficiency of adaptation to varying opponents in fighting games. Works evaluate whether adaptation occurs at all and if it leads to good performance, but do not focus on its efficiency.

Exhibition of human-like behavior is hard to compare due to the subjective nature of the evaluation. Nevertheless, we can note the success of imitation learning and case-based reasoning,

which are even the methods most opted for in commercial games. RL on the other hand tends to produce artificial behavior, even when mixing it with imitation learning. Although RL had less success, we can note hierarchical reinforcement learning and policy gradient methods produced more human-like behavior than other approaches within this paradigm, such as value-based methods. Still, many works on human likeness do not take into account the fairness of the agent, such as emulation of reaction time. The influence of a reaction time constraint on human-like behavior is yet to be determined.

There are also some key insights for the development of engaging artificial agents. For one, implementing various difficulty levels through versions of the agent at different stages of training should be handled with care, so that the undertrained agents are not close to unreasonable behavior. In addition, exploration of the environment by the agent should be guided rather than completely random. Another important takeaway is to generally avoid using ensembles of agents as a single agent since they can incur unnatural behavior shifts, which can disorient the human player. Developing populations of diverse agents is still underexplored, even though having competent and diverse opponents during training is important for improving performance, especially when applying self-play. For DDA, it is frequent to have the agent minimize the absolute difference in health between players, but it may produce unnatural behavior. Approaches using different objectives are still heavily unexplored, and care should be taken as the objectives themselves may not align with the player's expectations of an optimal game experience.

All works that attempted transfer learning corroborate its success in fighting games, both when considering transferability to playing as and against different characters.

There are still gaps in current research. The impact of some methods on achieving qualities of human-centered agents is yet to be studied, e.g., planning methods for generalizability.

VIII. FUTURE DIRECTIONS

In terms of environments, we can note that both DIAMBRA Arena and FightLadder have a collection of environments that are deserving of further study. Neither of them take reaction time into consideration, which is important for fighting games. The use of audio features for the development of agents is still underexplored, which is currently the focus of DareFightingICE.

In terms of performance, there is some nontransitivity between agent strategies in the literature. Performance is also hard to compare between approaches due to varied evaluation methodologies and opponents. As such, the research community can benefit from tournament-based benchmarks featuring various opponents, such as the one in [119]. The opponent selection in such a benchmark can also be used for population-based training, which proves to be useful for achieving good performance [30]. FightLadder is one such benchmark yet to be explored, focused on multiagent RL [30], although the selection of opponents is not as large as in [119] (5 versus 43) and was only trained on one of the provided games. Still, the benchmark lays a framework for evaluation of agents through four criteria, which can help standardize how agents are evaluated in the literature and make performance comparisons easier.

To the best of the authors' knowledge, there is a lack of works exploring quick adaptation of agents to different opponents in fighting games. At least in the realm of RL, it might be worth exploring model-based methods for few-shot learning, by learning a model of the environment. Learning an environment model may be alleviated since we can make use of the game itself as the environment model, such as the simulator in DareFightingICE, focusing merely on learning the opponent model. Model-free methods have the disadvantage of learning exclusively through experience; as such, the agent cannot learn from hypothetical situations. This contrasts with the way human players learn these games, which is by speculation. Fighting games usually provide modes for aiding learning. One is a "training mode" where players can manually set up specific gameplay scenarios to learn from them, without having to actually experience them against other players. Another is a "replay mode," where players can scrutinize past recorded situations to understand them better, and possibly even interact with them to see what would happen had each player acted differently. We may take inspiration from this, exploring model-based methods for efficient learning by focusing on uncertain or rarely experienced situations.

There is no standard method to measure adaptability. One possible way to determine adaptability is to measure how many time steps it takes for the agent to achieve a neutral result against the current opponent. This neutral result can be, for example, 50% win rate or an average health difference of zero as in [13]. Since the duration of time steps is standardized in these games, we can have a reliable measure of how long it takes an agent to adapt in real time. The metric being evaluated, either win rate or average health difference, requires sampling multiple games or time steps to be accurate, so a minimum amount of games or time steps needs to be enforced.

For RL, we can explore maximum entropy RL to build robust agents that can more reliably face arbitrary opponent strategies [120]. Robustness may therefore help mitigate the nontransitivity in agent strategies. In addition, we can attempt to use intrinsic reward formulations, such as "curiosity" [87], [121] to guide exploration and facilitate online learning of countermeasures against unexpected opponent strategies.

For human likeness, all works that addressed reaction time considered a constant (visual) reaction time. This is not entirely realistic, however, since it assumes *simple* reaction time. Rather, the prevalent form of reaction time in these games is *choice* reaction time. That is, instead of the player having to react to a single stimulus, it needs to react to multiple stimuli, with each one requiring different responses. The different stimuli constitute the different kinds of moves that the opponent can perform, and the player's expectations of which will be performed influence reaction time, as these games are heavily based on prediction. This means that the more uncertain the player is of the future, the greater the reaction time is. Ideally, the choice reaction time is the one that should be emulated, or else we are strictly dividing situations between those that a player can reliably react to and those that they cannot. In these games, there is nuance between actions that are easy and those that are hard to react to, rather than a hard threshold between those that can always be reacted to or not. Also, it can be argued that a 15-frame reaction time, corresponding to 250 ms and the one most commonly emulated,

is difficult to achieve by human players in practice. As such, current works are optimistic in the human abilities assumed for human-like artificial agents. However, it is not clear how choice reaction time can be emulated. There are different models for choice reaction time drawn from experimental data [49], [122], [123], but it is uncertain which model should be used. Since fighting games require sequential decision making, the work in [122] might be the most appropriate. Still, even with a model defined it is not obvious how reaction time should be modulated. It is possible to use uncertainty in an opponent model or environment model, or perhaps the breadth of a search tree in short-term planning, and even after reaction time is properly emulated, we have the issue of handling a variable reaction time, which has already been explored for RL [124], [125], [126]. This is a hard challenge when creating human-like agents using methods other than imitation-based ones, such as RL.

If we want to achieve fairness, then we need to address the fact that visual processing as done by human players affects decision making. It is possible to use visual ambiguity and occlusion as part of one's strategy, which affect the opponent's perception of the game state. When using structured representations of the game state, we are thus losing some fairness in this regard. Still, as long as the game has an appropriate sound design, these games may also be played through sound alone, so this factor loses some weight. Furthermore, parsing multidimensional input is usually done using convolutional neural networks, which are computationally expensive when applied to the ordinary hardware that players have access to. The importance of this factor can be debated, but we believe it is still worth mentioning.

For difficulty adjustment, we can explore creating agents for multiple difficulties by limiting their action space instead [52], which reduces the complexity of the game and makes it less overwhelming for newer players. Another possibility is to have the agent only execute the desired actions with a given probability, emulating the action execution difficulty of these games [52]. To determine player skill in DDA, we can make use of different metrics, such as those in [127]. Training multiple performant agents for different action space sizes or action execution probabilities is an interesting avenue for future work. We can also try to create easier agents by increasing the predictability of the agent's behavior, e.g., by lowering the temperature when using the softmax function for building action probability distributions.

Evolutionary algorithms are a great candidate for building large pools of diverse agents since these are a natural artifact of the algorithms [114], but they have still not been explored extensively. Quality-diversity algorithms are a class of evolutionary algorithms that aims to create populations of individuals that are not only fit but also diverse [128], and are yet to be explored in fighting games.

In general, there is yet to be a unified method that fulfills all qualities of a human-centered agent. We may combine the strengths of the methods presented in this work. For instance, we can employ SL to achieve human likeness, and RL to improve the agent's performance. Planning can be further incorporated to improve the agent's decisions, and evolutionary algorithms can be explored to create diverse agents. Finally, we may

perform some manual instrumentation on the agent, as rule-based methods provide predictability and allow handling exceptional cases, at the cost of generality.

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